Architecture of the Medical RAG System

1 Data Loading

Function:
Reads medical Q&A data from CSV files, labels them as [Patient] and [Doctor], and attaches department metadata.
Key Function:
load_medical_chunks()
Output: Clean text chunks + source paths

2 Embedding & Vector Database

Function:
Generates vector embeddings via Ollama (nomic-embed-text), stores them in an in-memory database with metadata.
Key Functions:
get_text_embedding(), build_vector_db()
Output: {text, embedding, metadata}

4 Answer Generation

Function:
Uses Llama3 model to synthesize a structured bilingual response based only on retrieved context.
Key Step: ollama.chat()
Output: Chinese explanation + English summary

Implementation developed at Wenzhou-Kean University

3 Retrieval

Function:
Computes cosine similarity between query and stored vectors; returns top-N relevant chunks above a similarity threshold (≥0.6).
Key Function: retrieve()
Output: Ranked list of relevant medical Q&A pairs

5 Interaction Layer

Function:
Handles user input, triggers retrieval and generation modules, and displays results in the terminal.
Key Function: run_chatbot()
Output: Continuous interactive Q&A loop